translocation analysis survival

2025-09-30

- load packages & read in processed dataset
- adjust ancestry so that *coastal = 0* and *inland = 1*

```
library(tidyverse)
library(NatParksPalettes)
library(viridis)
library(ggpubr)
library(readxl)
library(scico)

theme_set(theme_classic())

#manual change made to CJS output - deleted header lines to allow output to be read into R
cjs_out<-read.table("C:/Users/Steph/GitHub_data/translocation/CJS.beta.translocation.20250914.edit.txt"
           header=T)
cjs_out_pemberton<-read.table("C:/Users/Steph/GitHub_data/translocation/CJS.beta.translocation_Pemberton
           header=T)

phi<-read.csv("C:/Users/Steph/GitHub_data/translocation/CJS.fit.phi.translocation.20250914.csv")
phi_pemberton<-read.csv("C:/Users/Steph/GitHub_data/translocation/CJS.fit.phi.translocation_Pemberton.2

birds<-read_xlsx("C:/Users/Steph/TAMU_OneDrive/Thrushes/translocation/data_archiving/translocation_data
pembertonBirds<-read_xlsx("C:/Users/Steph/TAMU_OneDrive/Thrushes/translocation/data_archiving/transloca

birds2<-rbind(pembertonBirds,birds)%>%
  mutate(ancestry=hiest_ancestry*(-1)+1)

birds<-birds%>%
  mutate(ancestry=hiest_ancestry*(-1)+1)
```

```
tabS1<-birds2%>%
  mutate(ancestry=round(ancestry,3),
    capture_gps.n=round(capture_gps.n,3),
    capture_gps.w=round(capture_gps.w,3),
    release_gps.n=round(release_gps.n,3),
    release_gps.w=round(release_gps.w,3),
    release_site=if_else(release_site=="Lilloet","Lillooet",release_site))%>%
  dplyr::select(reference,translocated,ancestry,release_site,
    capture_gps.n,capture_gps.w,release_gps.n,
    release_gps.w,release_date)
#tabS1
#write_csv(tabS1,
#          "C:/Users/Steph/TAMU_OneDrive/Thrushes/translocation/fig_drafts/TableS1.csv")

rm(tabS1)
```

- Sample sizes for survival analysis

```
knitr::kable(birds2%>%
               group_by(release_site,release_year)%>%
               summarise(count=n()))
```

release_site	release_year	count
Lilloet	2020	20
Pemberton	2020	98
Pemberton	2023	90
Squamish	2023	30

CJS (mark recapture) models

- Fit a CJS model with detections binned into 10 day chunks to create 30 time points (data were too sparse with each day as a time point)
- Model: Survival ~ release_site + ancestry + release_site x ancestry; Detection ~ time + release_site + ancestry
- No evidence for the predicted interaction between release site and ancestry
- None of release site, ancestry, or the interaction between release site and ancestry had a significant effect on survival
- probability of detection varied among time points, with the higher probabilities of detection occurring during fall and spring migration
- birds released in Squamish had a higher probability of detection than those released in Lilloet, but detection probability did not vary with ancestry
- Once you control for difference in detection probability - no difference in survival between release sites

Survival analysis results:

```
knitr::kable(cjs_out%>%filter(startsWith(Factor,"Phi"))%>%
  mutate_if(is.numeric,round,digits=2)%>%
    select(-se)%>%
    rename(Lower_CL=lcl,Upper_CL=ucl))
```

Factor	Estimate	Lower_CL	Upper_CL
Phi.(Intercept)	0.57	-2.60	3.74
Phi.release_siteSquamish	1.19	-2.66	5.04
Phi.ancestry	3.55	-1.55	8.64
Phi.release_siteSquamish:ancestry	-1.39	-7.32	4.54

Detection probability analysis results:

(minus variation among time points)

```
knitr::kable(cjs_out%>%filter(startsWith(Factor,"p."))%>%
               filter(!startsWith(Factor,"p.time"))%>%
  mutate_if(is.numeric,round,digits=2)%>%
    select(-se)%>%
    rename(Lower_CL=lcl,Upper_CL=ucl))
```

Factor	Estimate	Lower_CL	Upper_CL
p.(Intercept)	-0.89	-2.51	0.74
p.release_siteSquamish	2.81	1.94	3.67
p.ancestry	-0.33	-2.52	1.86

CJS - with Pemberton

- Here, I would expect a flat relationship between ancestry and survival in Pemberton (ie at the centre of the hybrid zone), a positive relationship in Lilloet (better survival of inland birds) and a negative relationship in Squamish (better survival of coastal birds)
- Model: Survival ~ release_site + ancestry + release_site x ancestry; Detection ~ time + release_site + release_year + ancestry
- Adding Pemberton birds tagged in 2020 & 2023 does not change anything for the survival results - everything is still null
- Because release site and release year aren't entirely covarying, I added relase year (alongside time point, release site, and ancestry) as a predictor for detection probability

Survival analysis results:

```
knitr::kable(cjs_out_pemberton%>%filter(startsWith(Factor,"Phi"))%>%
  mutate_if(is.numeric,round,digits=2)%>%
    select(-se)%>%
    rename(Lower_CL=lcl,Upper_CL=ucl))
```

Factor	Estimate	Lower_CL	Upper_CL
Phi.(Intercept)	2.97	2.26	3.67
Phi.release_siteLilloet	-2.05	-4.76	0.66
Phi.release_siteSquamish	-0.95	-3.43	1.54
Phi.ancestry	0.10	-0.98	1.18
Phi.release_siteLilloet:ancestry	2.71	-1.32	6.74
Phi.release_siteSquamish:ancestry	1.61	-2.03	5.26

Detection probability analysis results:

```
knitr::kable(cjs_out_pemberton%>%filter(startsWith(Factor,"p."))%>%
               filter(!startsWith(Factor,"p.time"))%>%
  mutate_if(is.numeric,round,digits=2)%>%
    select(-se)%>%
    rename(Lower_CL=lcl,Upper_CL=ucl))
```

Factor	Estimate	Lower_CL	Upper_CL
p.(Intercept)	-0.27	-0.82	0.27
p.release_siteLilloet	-1.09	-1.80	-0.38
p.release_siteSquamish	0.35	0.02	0.67
p.release_year2023	1.14	0.86	1.42
p.ancestry	0.25	-0.49	0.98

```r
library(lme4)
library(car)

glm1<-glm(survival~ancestry*release_site,
      family = binomial(link="logit"),
      data=birds)

summary(glm1)
```

```
##
## Call:
## glm(formula = survival ~ ancestry * release_site, family = binomial(link = "logit"),
##     data = birds)
##
## Coefficients:
##                                Estimate Std. Error z value Pr(>|z|)
## (Intercept)                      -5.120      2.994  -1.710   0.0872 .
## ancestry                          5.073      3.988   1.272   0.2034
## release_siteSquamish              4.084      3.478   1.174   0.2402
## ancestry:release_siteSquamish    -4.141      4.731  -0.875   0.3815
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 61.086  on 49  degrees of freedom
## Residual deviance: 54.880  on 46  degrees of freedom
## AIC: 62.88
##
## Number of Fisher Scoring iterations: 5
```

```r
Anova(glm1)
```

```
## Analysis of Deviance Table (Type II tests)
##
## Response: survival
##                       LR Chisq Df Pr(>Chisq)
## ancestry                1.5475  1    0.21350
## release_site            2.9227  1    0.08734 .
## ancestry:release_site   0.8618  1    0.35323
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
glmm1<-glmer(survival~ancestry*release_site+(1|release_year),
      family = binomial(link="logit"),
      data=birds2)

summary(glmm1)
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
##   Approximation) [glmerMod]
```

```
##  Family: binomial  ( logit )
## Formula: survival ~ ancestry * release_site + (1 | release_year)
##    Data: birds2
##
##      AIC      BIC   logLik deviance df.resid
##    278.1    302.4   -132.0    264.1      231
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -0.9255 -0.6497 -0.5059  1.1647  2.5262
##
## Random effects:
##  Groups       Name        Variance Std.Dev.
##  release_year (Intercept) 0.1143   0.3381
## Number of obs: 238, groups:  release_year, 2
##
## Fixed effects:
##                                  Estimate Std. Error z value Pr(>|z|)
## (Intercept)                        -4.864      3.010  -1.616    0.106
## ancestry                            5.098      3.991   1.277    0.202
## release_sitePemberton               3.304      3.075   1.075    0.283
## release_siteSquamish                3.535      3.503   1.009    0.313
## ancestry:release_sitePemberton     -4.397      4.120  -1.067    0.286
## ancestry:release_siteSquamish      -4.160      4.738  -0.878    0.380
##
## Correlation of Fixed Effects:
##             (Intr) ancstr rls_sP rls_sS anc:_P
## ancestry    -0.970
## rls_stPmbrt -0.973  0.949
## rls_stSqmsh -0.858  0.833  0.839
## ancstry:r_P  0.939 -0.969 -0.972 -0.806
## ancstry:r_S  0.817 -0.842 -0.800 -0.969  0.816
```

```r
Anova(glmm1)
```

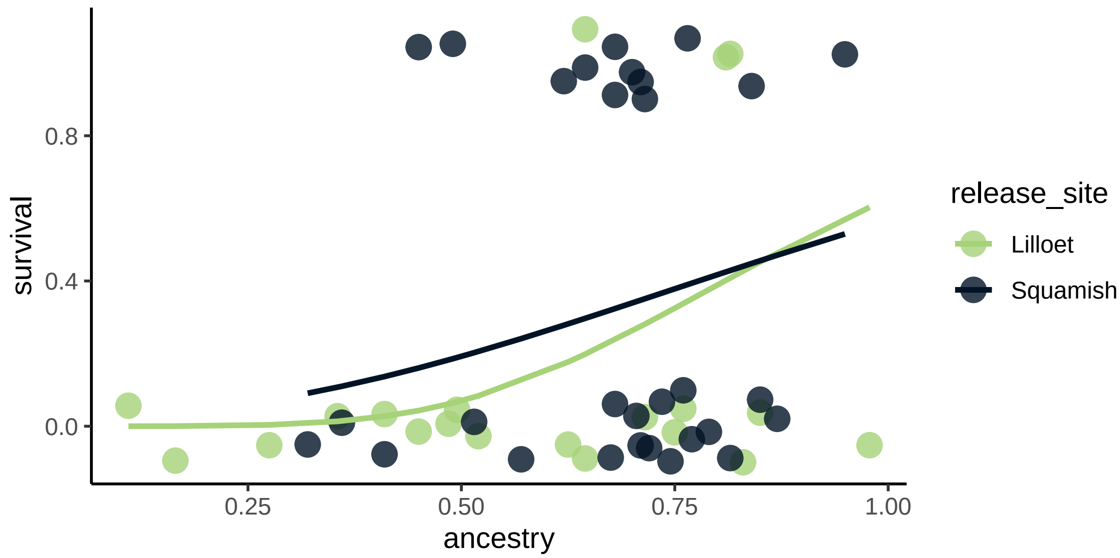
```
## Analysis of Deviance Table (Type II Wald chisquare tests)
##
## Response: survival
##                        Chisq Df Pr(>Chisq)
## ancestry              1.0976  1     0.2948
## release_site          0.7466  2     0.6885
## ancestry:release_site 1.1396  2     0.5656
```

Survival figures

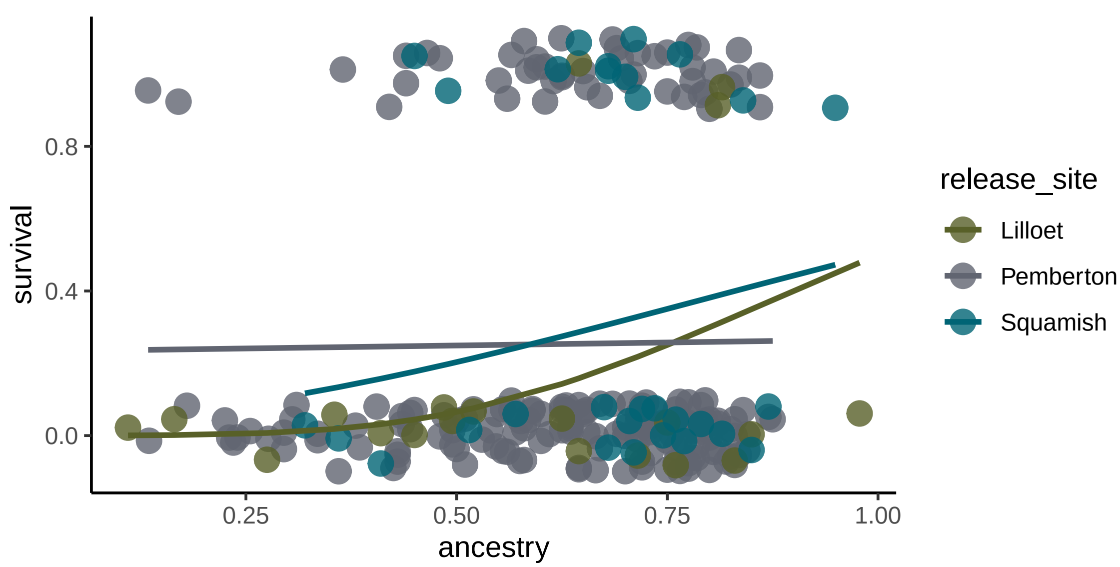
- slopes are based on ancestry x release site predictions from the CJS models
- I think I actually prefer the violin plots because the data are more visible

```r
ggplot(birds,aes(x=ancestry,y=survival,
                 group=release_site,
                 colour=release_site))+
  #geom_smooth(method="lm",alpha=0.18)+
```

```
  geom_point(size=4,shape=19,alpha=0.8,position=position_jitter(width=0,height=0.1))+
  geom_line(data=phi,aes(x=ancestry,y=est.adj,colour=release_site),size=1)+
  scale_color_manual(values=scico(n=3,palette='navia',end=0.8)[c(3,1)])+
  xlab("ancestry")+
  ylab("survival")
```

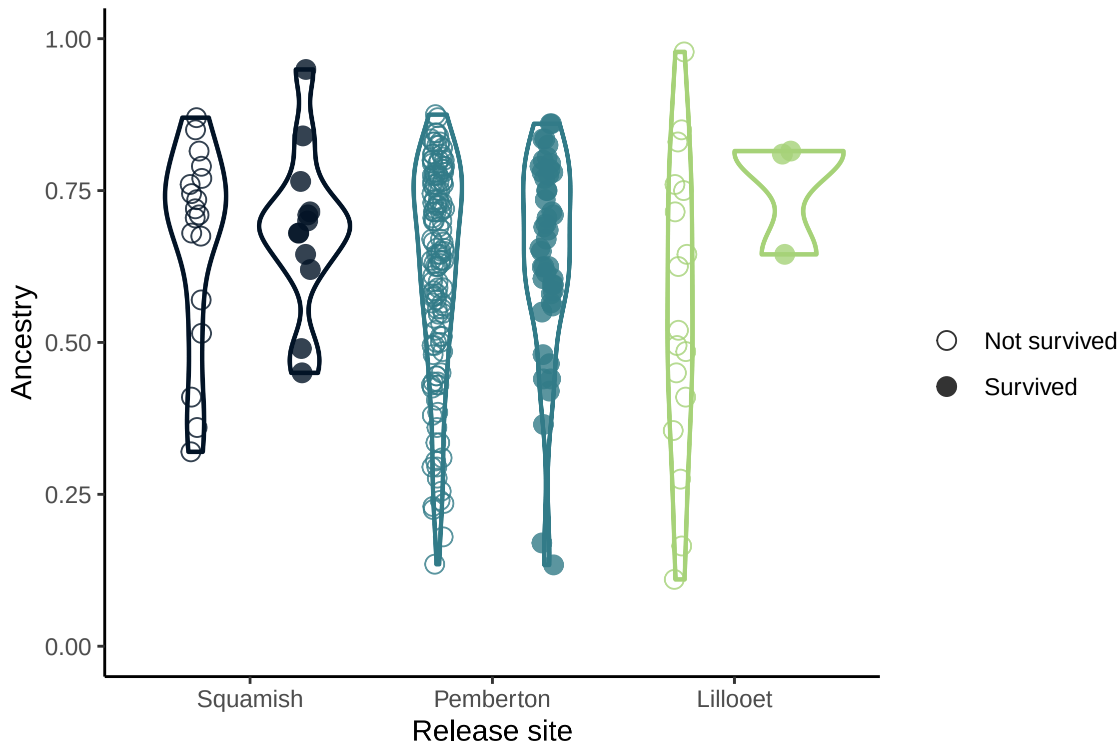



```
ggplot(birds2,aes(x=ancestry,y=survival,
                 group=release_site,
                 colour=release_site))+
  geom_point(size=4,shape=19,alpha=0.8,position=position_jitter(width=0,height=0.1))+
  geom_line(data=phi_pemberton,
          aes(x=ancestry,y=est.adj,colour=release_site),size=1)+
  scale_color_manual(values=natparks.pals("Banff")[c(4,6,1)])+
  xlab("ancestry")+
  ylab("survival")
```



```r
birds2$release_site<-factor(birds2$release_site,levels=c("Squamish","Pemberton","Lilloet"))
```

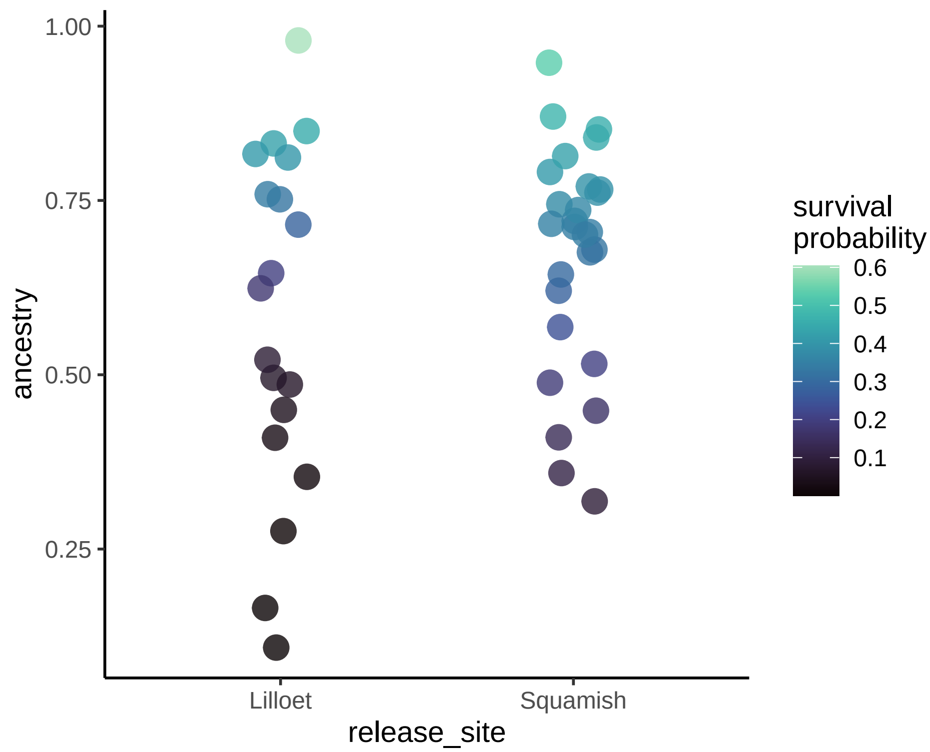
```r
gg1<-ggplot(birds2,aes(y=ancestry,x=release_site,
                 group=interaction(as.factor(survival),release_site),
                 colour=release_site,shape=as.factor(survival)))+
  geom_point(size=3,alpha=0.8,
           position=position_jitterdodge(
             jitter.width =0.2,jitter.height=0,dodge.width=0.9))+
  scale_shape_manual(values=c(1,19),name="",labels=c("Not survived","Survived"))+
  scale_color_manual(values=scico(n=3,palette='navia',end=0.8),
                     guide="none")+
  geom_violin(fill=NA,aes(group=interaction(survival,release_site)),size=0.8)+
  ylab("Ancestry")+xlab("Release site")+ylim(0,1)+
  scale_x_discrete(labels=c("Squamish","Pemberton","Lillooet"))

gg1
```




```r
#ggsave("C:/Users/Steph/TAMU_OneDrive/Thrushes/translocation/Fig2.svg",
#       plot=gg1,width=6,height=4)
```

- Next, points coloured by model-fitted survival probabilities

```r
ggplot(phi,aes(x=release_site,y=ancestry,colour=est.adj))+
  geom_jitter(size=4,shape=19,alpha=0.8,width=0.1)+
  viridis::scale_colour_viridis(option="mako",end=0.9,name="survival\nprobability")
```




```
ggplot(phi_pemberton,aes(x=release_site,y=ancestry,colour=est.adj))+
  geom_jitter(size=4,shape=19,alpha=0.8,width=0.2)+
  viridis::scale_colour_viridis(option="mako",end=0.9,name="survival\nprobability")
```